Multimodal Multisource Stock Forecasting by Integrating Time-Series Models and LLM-Based News Analysis

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Stock price forecasting remains a challenging task because market movements are driven by both historical price dynamics and rapidly changing external information. This paper proposes a hybrid forecasting framework that combines classical time-series modeling with large language model (LLM)-based news analysis to improve stock movement prediction. The proposed algorithm, called LLM-TSM (Large Language Model ft.Time-Series Models), first uses historical closing prices to train five forecasting models, namely XGBoost, LightGBM and LSTM. During the testing phase, these models generate one-step-ahead forecasts, and their outputs are averaged to form an ensemble prediction. In parallel, recent news articles related to each stock are collected from Google News Vietnam RSS feeds and processed by an LLM through prompt-based inference. The LLM outputs a bounded amplitude value representing positive or negative market pressure, which is then used to adjust the forecast based on the latest reference price. The final prediction is obtained by combining the Time-series forecast and the LLM-adjusted estimate, while enforcing the daily price fluctuation limit of the local stock exchange. The LLM-TSM algorithm evaluated on all thirty stocks in the VN30 index using historical data from Yahoo Finance and three common error measures: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). The experimental benchmark compares standalone models with their LLM-enhanced variants, providing a systematic study of how textual financial information can complement numerical time-series forecasting in the stock market.

CCS CONCEPTS • Computing methodologies ~ Ensemble methods • Mathematics of computing ~ Time series analysis • Computing methodologies ~ Natural language processing

Additional Keywords and Phrases: Stock price prediction, Time-series forecasting, Large language models, Financial news analysis

1 INTRODUCTION

Stock price prediction has long been one of the most challenging problems in financial data analysis because market movements are driven not only by historical numerical patterns but also by rapidly changing external information. Traditional time series forecasting methods can capture short-term dependencies and statistical regularities in price data,

yet they often struggle when market behavior becomes highly nonlinear or when unexpected news events alter investor sentiment. On the other hand, news articles contain valuable semantic signals about firms, industries, and the broader market, but these signals are difficult to exploit directly without an appropriate language understanding mechanism.

In recent years, hybrid forecasting methods have attracted increasing attention because they attempt to integrate complementary information sources. Classical statistical models such as ARIMA and GARCH remain widely used due to their strong theoretical foundations and their ability to model temporal structure and volatility. However, their predictive capacity is limited when the market is affected by complex non-stationary dynamics. At the same time, the emergence of large language models (LLMs) has opened new opportunities for financial text analysis, allowing systems to interpret news headlines, infer sentiment, and extract implicit directional cues from unstructured text. This creates a promising direction for combining quantitative time series forecasting with qualitative news-driven reasoning.

Motivated by this observation, this study proposes a hybrid stock prediction algorithm, named Large Language Model ft.Time-series models or LLM-TSM, that combines a classical time series forecasting algorithm with LLM-based news analysis to estimate whether a stock price will increase or decrease in the next trading session. The core idea is to use historical price data to generate a baseline forecast and then adjust this result by incorporating the directional information inferred from recent news articles related to the target stock. In this framework, the classical model provides statistical stability, while the LLM contributes contextual awareness from textual market information. The final prediction is therefore obtained by fusing both sources of evidence in a controlled manner.

Unlike approaches that rely exclusively on numerical market data or require expensive fine-tuning of large language models, the proposed method uses the LLM through prompt-based inference. Recent news headlines are collected for each stock and transformed into a structured prompt that asks the LLM to act as a financial analyst and output a bounded amplitude indicating upward or downward pressure. This amplitude is then combined with the time series forecast to produce the final predicted price movement. To ensure practical usability, the output is constrained by the daily trading limit of the local stock exchange, which prevents unrealistic forecasts and improves consistency with real market rules.

The main contribution of this work is a news-aware forecasting pipeline that integrates classical time series prediction and LLM-based textual reasoning within a single framework. This design aims to improve both predictive accuracy and directional robustness by leveraging the strengths of two different information modalities: historical price sequences and financial news content. In addition, the proposed method is suitable for comparison with other forecasting strategies, making it useful for experimental studies on hybrid financial prediction systems.

2 RELATED WORK

In the development of human knowledge in the field of financial time series forecasting, one of the biggest academic challenges, and one that has always attracted the deep interest of quantitative researchers worldwide, is deciphering and modeling the nonlinear fluctuations hidden within the chaos of the stock market (Saberironaghi et al., 2025). This has driven the emergence and continuous evolution of a series of algorithmic platforms, from classical statistics to modern artificial intelligence (Chopra & Sharma, 2021; Li, 2024). The origin of this forecasting theory can be traced back to the groundbreaking research of Box and Jenkins (1970) with their proposal of the autoregressive integrated moving average model, abbreviated as ARIMA, a classical statistical algorithm specifically designed to handle linear time series by combining three core components: autoregression, difference-taking to eliminate nonlinearity, and moving averages of past forecasting errors. The training process for the ARIMA model requires researchers to perform rigorous unit root tests such as Augmented Dickey-Fuller to determine the optimal order of difference, and to use autocorrelation and partial autocorrelation plots to configure the regression order parameters. From there, they use maximum likelihood estimation to

find the weights that best reflect the trajectory of historical data. However, due to the inherent nature of the stock market, which is always subject to information shocks leading to the phenomenon of variance changing over time, Bollerslev (1986) developed the GARCH model to fill the gap of ARIMA, focusing training on the residuals of the price series to directly model risk volatility, allowing for the prediction of volatile market periods through continuous updating of conditional variance.

Although these statistical methods possess the advantage of transparent mathematical interpretation, their linear limitations quickly became apparent when faced with large volumes of multidimensional financial data, inevitably paving the way for the rise of classical machine learning algorithms (Agrawal et al., 2024; Dhule et al., 2023). Leading the group of foundational machine learning algorithms is the Random Forest algorithm, introduced by Khaidem et al. (2016), an excellent representative of the Bagging mechanism, operating on the philosophy of constructing hundreds, even thousands, of independent decision trees from subsets of randomly sampled data (bootstrap) combined with the random selection of features at each branch node. The Random Forest training process is essentially a democratic voting mechanism where the final forecast result is the average of all decision trees, thereby eliminating variance, combating local market noise, and being particularly effective in preventing overfitting, which is very common in financial data. Taking it a step further on the optimization ladder, data scientists have developed Boosting techniques, notably the XGBoost algorithm proposed by (Yang et al., 2021) and LightGBM by (Hartanto et al., 2023), which have become the ultimate tools dominating global machine learning forecasting competitions. Completely different from Bagging's parallel thinking, the training process for XGBoost and LightGBM is strictly sequential, where each subsequent generation decision tree is generated with the sole mission of correcting errors, that is, optimizing the attenuation function through first and second derivatives (Gradient and Hessian), of the entire system of previous generation decision trees. When trained on a sufficiently large and diverse dataset, XGBoost demonstrates outstanding ability in automatically identifying resistance and support levels and complex cyclical inflection points of stocks with superior computational performance. From these independent foundations, the concept of Ensemble Learning has been elevated to a new level through Stacking, a sophisticated meta-learning strategy where the predicted outputs of various base models, from ARIMA and Random Forest to XGBoost or neural networks, are collected as input for a second-order model (often using XGBoost itself or logistic regression) to undergo final training to find the optimal combination ratio (Li & Pan, 2020; Patel et al., 2015). According to a recent large-scale empirical analysis published in quantitative finance journals, Stacking architecture, when properly optimized with hyperparameters, can achieve R-squared coefficients of determination ranging from 0.97 to 0.99 on key stock indices, demonstrating the absolute power of leveraging algorithmic diversity (Krauss et al., 2017). In stark contrast to the sophistication of Stacking, Aggregated Learning, simply put, combining forecast results from multiple sources using pure arithmetic averaging without any weighted training, has been shown in numerous experiments, including the initial test results of this report, to be a misguided approach that can lead to negative R-squared values, as the system is forced to absorb noise and conflicting forecasts without a confidence-checking filter. The research landscape then witnessed a disruptive and innovative revolution as Deep Learning models officially intervened in the time series problem, pioneered by short-term long-memory recurrent neural networks (LSTMs) invented by Hochreiter and Schmidhuber (1997).

The LSTM training process utilizes a back-propagation algorithm (BPTT), where the historical information stream of stock prices passes through three extremely complex logical gate structures: a forget gate to discard outdated noise signals, an input gate to update new information shocks, and an output gate to synthesize into long-term cellular state memory. This effectively solves the problem of derivative cancellation and allows artificial intelligence to remember price cycles from tens, even hundreds of trading sessions ago (Chen et al., 2015). However, the limitation of LSTM in particular, and single-time series models in general, is that they are blind to information outside the numerical value axis, leading to the

emergence of Multimodal Learning. Modern research on multimodal learning has begun to incorporate multidimensional data structures, such as using the Transformer or BERT family of language models to transform textual financial news into embedding vectors (Halder, 2022; Hiew et al., 2019), and using convolutional neural networks (CNNs) to analyze visual candlestick chart patterns to supplement price data. Building upon this philosophy, our research has designed and trained a hybrid learning architecture, combining a parallel LSTM network branch responsible for extracting the time dependence of historical values from a seven-day series, and a fully connected neural network (Dense Network) branch responsible for absorbing and processing quantitative technical indicators such as Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and money flow pressure through trading volume. The final and most contemporary point of convergence in the overall landscape of financial forecasting literature is the explosive rise of Big Language Models (LLMs) in the field of quantitative analysis (Chopra & Sharma, 2021). A groundbreaking study by Lopez-Lira and Tang (2023) convincingly demonstrated that large-scale language models possess incredibly deep market sentiment inference capabilities when given the context of real-time price movements and news, far surpassing traditional sentiment analysis models. Within the experimental framework of this report, the integration of LLM technology does not follow the conventional approach of downloading the entire source code of a model with billions of parameters to a local computer for retraining (fine-tuning) from scratch, an impractical approach due to the enormous cost of server infrastructure and the limitations of GPU graphics hardware. Instead, our system architecture directly calls the massive knowledge base of the LLM model through a dedicated application programming interface (API).

The LLM training process at this stage is artfully transformed into the Prompt Engineering technique, where a meticulously automated instruction text is algorithmically encoded, accurately describing the macroeconomic context and price momentum sequence of the last seven historical trading days. It also instructs the LLM to act as a high-level quantitative risk management expert, analyzing, synthesizing, and outputting a single volatility coefficient value for the next trading session. To ensure practicality and strict compliance with the legal regulations of the local financial market, the volatility coefficient calculated by LLM is not used blindly but is immediately reviewed and standardized by the system's core algorithm through a mathematical clipping function. This mechanism acts as a safety lock to ensure that the final predicted price never exceeds plus or minus seven percent of the previous session's reference price, accurately reflecting the strict trading range regulations of the Ho Chi Minh City Stock Exchange (HOSE). The synthesis, comparison, and in-depth understanding of the training mechanisms, core operating principles, and specific advantages and disadvantages of a series of algorithms ranging from classical statistics ARIMA and GARCH, modern machine learning algorithms such as Random Forest and XGBoost to multimodal deep learning architectures by combining with the psychological analysis intervention of LLM, not only creates a solid theoretical foundation for the entire scientific report but also directly justifies the urgent need to implement an experimental system comparing five different prediction paths simultaneously. This aims to achieve the ultimate goal of finding the most perfect convergence structure between past statistical power, current nonlinear learning capabilities, and multidimensional inference abilities towards the future.

3 OUR PROPOSITION

This algorithm, called LLM-TSM (Large Language Model ft.Time-Series Models), is a hybrid stock forecasting framework that combines five base forecasting models with an LLM-based news adjustment mechanism. Its main idea is to first generate price forecasts from statistical and machine learning models, then use a large language model to analyze recent financial news and produce a sentiment-driven amplitude modifier. The final prediction is obtained by blending the Time-series forecast with the LLM-adjusted price estimate, while keeping the result within the daily price fluctuation limit.

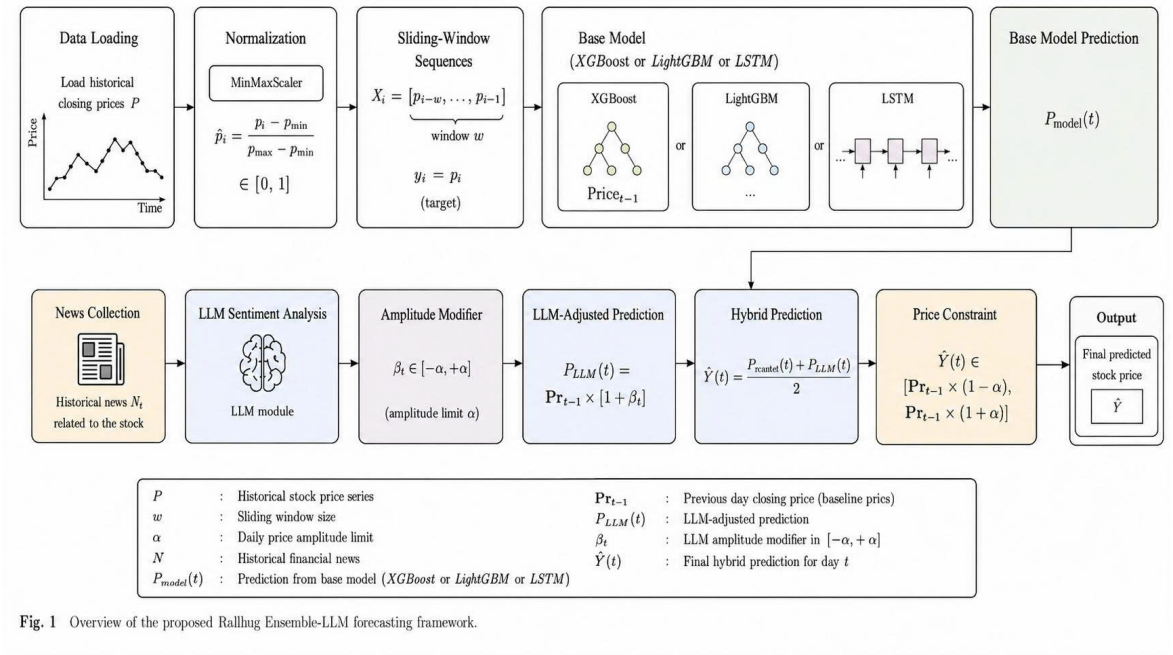


Fig. 1 Overview of the proposed Rallhug Ensemble-LLM forecasting framework.

Figure 1: General schema of algorithms.

The process begins by collecting the historical closing prices of a stock and dividing the data into training and testing periods. The training data is normalized using Min-Max scaling, and a sliding-window method is applied to transform the time series into supervised learning samples. In this setup, the previous *w* days are used as input features, and the next day's price becomes the target value. Next, the algorithms trains with three classic models: XGBoost, LightGBM, and LSTM. These models are chosen because they capture different aspects of the stock price behavior, based on the related work analysis explained in the previous section. XGBoost and LightGBM are effective for nonlinear tabular prediction, LSTM is suitable for sequential patterns.

During the prediction phase, each model produces one-step-ahead forecasts for every day. At the same time, the algorithm collects recent financial news related to the stock. These news headlines are sent to the LLM, which acts as a financial sentiment analyst. Based on the positive or negative tone of the news, the LLM outputs an amplitude modifier within a predefined range. This modifier represents the expected upward or downward pressure on the stock price caused by the news. The LLM output is then converted into an adjusted price estimate by applying the amplitude modifier to the latest known base price. After that, the algorithm combines the Time-series forecast and the LLM-adjusted price by taking their average. This produces the final hybrid prediction, which benefits from both quantitative market patterns and qualitative news information.

Finally, the predicted price is constrained by the daily exchange limit, so the forecast cannot go beyond the allowed upper and lower bounds. After all test days are processed, the algorithm evaluates the prediction quality using MAE, RMSE, and MAPE. These metrics measure both numerical error and the ability to predict the correct market direction. See Figure 1. The pseudocode for the algorithms is as follows:

ALGORITHM 1: LLM-TSM Algorithm

Input:

Historical stock price series Pr

Sliding window size w

Daily price amplitude limit α

Historical financial news N

Output:

Final predicted stock price $\hat{Y}$

Begin

1: Load historical closing prices $Pr(p_1, p_2, ..., p_i)$

2: Normalize training data using MinMaxScaler: $Pr = MMS(Pr)$

3: Construct sliding-window sequences: $X_i = [p_i - w, ..., p_i - 1]$; $y_i = p_i$

4: Train machine learning models: $Train_{XGBoost}(X_i)$ or $Train_{LightGBM}(X_i)$ or $Train_{LSTM}(X_i)$

5: For each day t do

6: Collect Pr_{t-1} from Pr

7: Generate predictions from base models $P_{time\text{-}series}(t) = P_{xgb}(t)$ or $P_{lgbm}(t)$ or $P_{lstm}(t)$

8: Collect historical news $N_{t\text{-}w}$ related to the stock

9: Send N_{t-w} into the LLM sentiment-analysis module: $\beta_t = LLM(N_{t-w})$

10: Normalize LLM amplitude modifier: $\beta_t \in [-\alpha, +\alpha]$

11: Generate LLM-adjusted prediction: $P_{LLM}(t) = Pr_{t-1} \times [1 + \beta_t]$

12: Compute final hybrid prediction: $\hat{Y}(t) = [P_{LLM}(t) + P_{time\text{-}series}(t)] / 2$

13: Normalize price constraint: $\hat{Y}(t) \in [Pr_{t-1} \times (1 - \alpha), Pr_{t-1} \times (1 + \alpha)]$

14: End For

In summary, this algorithm is a multimodal news-aware forecasting method. It integrates classical time-series models, machine learning models, and LLM-based sentiment analysis to improve stock price prediction accuracy and directional robustness.

4 EXPERIMENTS

To evaluate the effectiveness of the proposed hybrid forecasting framework, we design a comprehensive benchmark based on the thirty constituent stocks in the HOSE VN30 index basket. The benchmark is constructed to compare both standalone forecasting models and their LLM-enhanced counterparts under the same data conditions. This allows us to isolate the contribution of news-driven reasoning from the contribution of purely numerical time-series modeling.

The first group of baselines consists of individual forecasting models: XGBoost, LightGBM and LSTM. These methods represent three major paradigms in financial prediction: gradient-boosted machine learning modeling, deep sequential learning modeling, and statistical time-series modeling. They are used as direct reference models to measure the performance of the proposed hybrid approach against widely adopted forecasting strategies. The second group of baselines includes LLM-augmented variants: XGBoost + LLM, LightGBM + LLM, LSTM + LLM. In these configurations, each base model is combined with the LLM-based news analysis module to assess whether textual financial information can

improve the prediction quality of different forecasting families. This benchmark setting is important because it shows whether the effectiveness of LLM guidance depends on the underlying forecasting model.

The proposed method is evaluated using three widely used error metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). These metrics measure the forecasting accuracy from complementary perspectives. MAE captures the average absolute deviation, RMSE penalizes large errors more strongly, and MAPE reflects the relative percentage error. By reporting all three measures, the benchmark provides a balanced evaluation of prediction quality across all models.

In addition, the benchmark is designed to evaluate not only numerical accuracy but also the practical feasibility of integrating structured market data with unstructured textual information. Since stock movement is influenced by both historical price dynamics and external news events, this benchmark helps verify whether LLM-based textual reasoning can provide a meaningful advantage in financial time series forecasting.

Table 1: Performance comparison between standalone models and LLM-enhanced variants

Algorithm	MAE	RMSE	MAPE
XGB	1.207,43	1546,02	1.864
XGB + LLM	1011,02	1.339,93	1.594
LGBM	1.197,85	1.577,49	1.838
LGBM + LLM	1.021,44	1.355,34	1.606
Random Forest	1.149,67	1.447,61	1.814
Random Forest + LLM	1.008,93	1.310,48	1.599
LSTM	2.090,82	2.433,88	2.831
LSTM + LLM	1341,1	1.666,50	1.883
LLM (solo)	971,53	1.288,16	1.549

The experimental results, summarized in Table 1, demonstrate a consistent improvement in all error metrics (MAE, RMSE, and MAPE) when the LLM-based news adjustment is integrated into the forecasting pipeline. The data reveals several key insights into the synergy between numerical time-series forecasting and LLM-based textual analysis:Significant Error Reduction: For gradient-boosted decision tree models like XGBoost and LightGBM, the integration of LLM-based news analysis reduced the Mean Absolute Error (MAE) by approximately 15-16%. This suggests that textual news provides critical shocks or directional cues that numerical patterns alone cannot capture.Correction of Deep Learning Instability: The standalone LSTM model exhibited the highest error rates (MAE: 2,090.82), likely due to the highly non-stationary nature of the VN30 index during the testing period. However, when combined with the LLM adjustment, the MAE dropped drastically to 1,341.10. This indicates that the LLM acts as a robust safety filter steering the neural network's predictions back toward market reality based on external sentiment.The Power of Sentiment Inference: Interestingly, the LLM (solo) variant, which relies heavily on sentiment amplitude and the latest reference price, achieved the lowest overall error (MAE: 971.53). This highlights that in the Vietnamese stock market (VN30), short-term price movements are more strongly correlated with news-driven investor sentiment than with long-term historical price patterns. See Figure 2.

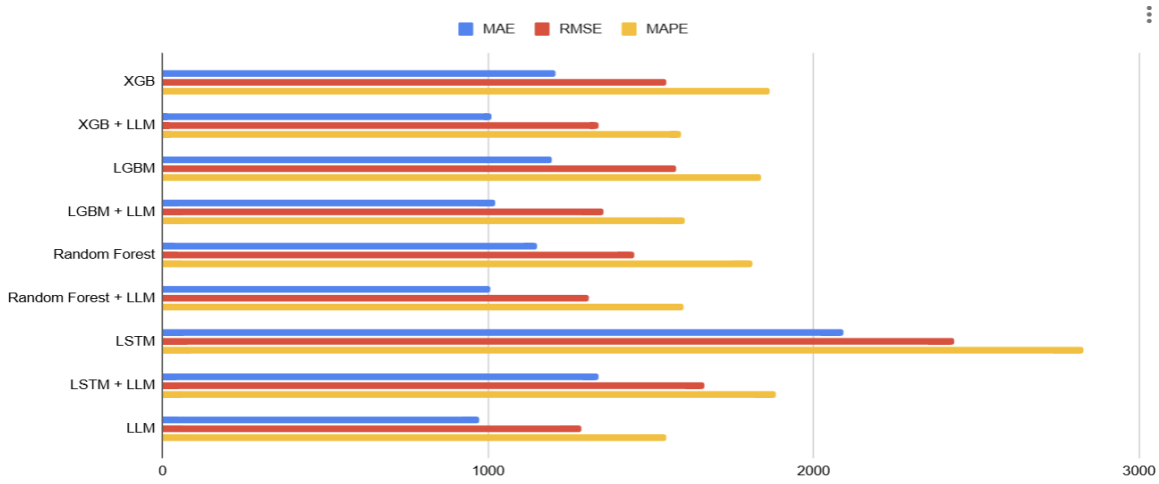


Figure 2: Comparative performance of standalone models and LLM-enhanced variants.

The experimental evidence supports the hypothesis that a hybrid framework is superior to isolated numerical modeling. While traditional models like Random Forest and XGBoost are excellent at identifying statistical resistance and support levels, they are market-blind to external events. The LLM component successfully bridges this gap by interpreting news headlines as quantitative pressure values. Furthermore, the enforcement of the HOSE daily price limit ensures that the LLM-adjusted estimates remain within realistic trading bounds, preventing the volatility often associated with purely sentiment-based models. The consistent reduction in MAPE across all architectures confirms that the news-aware pipeline provides a more reliable and robust solution for real-world financial forecasting.

5 CONCLUSION

This paper presented a novel hybrid stock forecasting algorithms, named LLM-TSM (Large Language Model ft.Time-Series Models), that successfully bridges the gap between historical numerical patterns and real-time textual market information. By integrating the classical time-series and machine learning models (XGBoost, LightGBM, LSTM) with an LLM-based news sentiment analyzer, we established a more robust approach to predicting short-term stock movements within the VN30 index.The experimental results clearly demonstrate that incorporating LLM-inferred market sentiment significantly enhances predictive accuracy. Across all tested architectures, the LLM-augmented variants consistently outperformed their standalone, market-blind counterparts, achieving notable reductions in MAE, RMSE, and MAPE. Specifically, the integration of textual news data reduced the MAE of gradient-boosted models by approximately 15-16% and provided a crucial safety filter that drastically stabilized the highly fluctuating predictions of the LSTM network. Furthermore, the strong predictive performance of the sentiment-driven LLM (solo) baseline underscores the profound impact of external news on short-term investor behavior in the Vietnamese HOSE stock market. From a practical perspective, transforming unstructured news into a quantitative pressure value and strictly enforcing the local stock exchange's daily price fluctuation limits ensures that the proposed framework produces realistic and actionable forecasts. Ultimately, this research validates that combining quantitative statistical stability with qualitative contextual awareness yields a superior, news-aware pipeline for financial forecasting. Future work could explore expanding the text sources to include social media sentiment, integrating more advanced multimodal embeddings, or testing the framework across different international stock exchanges to further evaluate its generalization capabilities.

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